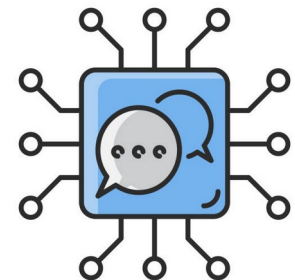


Language Analysis and Computational Linguistics

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- What is Language Analysis?
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- Natural Language Generation (NLG)
- Natural Language Interaction (NLI)

Session 1:

Introduction

What is Language Analysis?

- It is the systematic process of deconstructing language to understand its structure, components, and principles.
- It's how linguists (and NLP engineers) build a "blueprint" of a language.
- It is a descriptive science, not a prescriptive one.
 - Prescriptive (Not our goal): "You should not end a sentence with a preposition." (e.g., grammar school)
 - Descriptive (Our goal): "A large percentage of English speakers do end sentences with prepositions. What is the rule?"

Saussure's Two Axes

- Ferdinand de Saussure (a key linguist) proposed that language operates along two fundamental axes.
 - 1. Syntagmatic Relationship (The "Chain"):
How units combine in sequence.
 - 2. Paradigmatic Relationship (The "Choice"):
How units can substitute for one another.
- Understanding this is the key to understanding all language structure, from n-grams to syntax trees.

The Syntagmatic Axis (The "Chain")

- Concerns the linear sequence of words. It's a relationship of "AND".
- The AND cat AND sat...
- The rules of syntax and morphology are primarily syntagmatic.
- Example: "The cat sat" is valid. "Cat the sat" is not.
- Computational Link: This is the foundation of n-grams ((bigram: "the cat")), Part-of-Speech (PoS) tagging, and syntactic parsing.

The Paradigmatic Axis (The "Choice")

- Concerns the substitution of words in a "slot". It's a relationship of "OR".
 - The [cat | dog | man | student] sat.
- Words on this axis share a grammatical category (e.g., Nouns) or a semantic field.
- Computational Link: This is the foundation of word embeddings (Word2Vec, GloVe), where similar words (e.g., "cat", "dog") are "close" in vector space. Also key for synonym replacement and Word Sense Disambiguation (WSD).

The Three Lenses of Analysis

- When we analyze any part of language (a word, a sentence), we must ask three separate questions:
- 1. Form: What does it look or sound like? (The string of letters, the sound waves).
- 2. Function: What job is it doing in the sentence? (Its grammatical role).
- 3. Meaning: What concept does it refer to? (Its semantics).
- Crucial for NLP: Ambiguity arises when one form has multiple functions or meanings.

What is Paradigm?

- If we take another look at the example sentences, we used for syntagms,
Anne killed a mosquito.
- Here, mosquito can be replaced by a variety of words that would imply similar meanings.
- Anne might have killed a spider, bug, ant or another insect. Each of these words belongs to a paradigm of animals or insects Anne might have killed.

What is Syntagm?

- Syntagm is a linguistic unit which consists of a set of linguist forms such as letters, phonemes or words that are in a sequential relationship to one another.
- A syntagm is also known as a chain of signifiers.
- The relationship within a syntagm is known as a syntagmatic relationship.
- Syntagmatic relationship involves a sequence of signs that create meaning as a whole. They are all about the positioning.

What is Syntagm?

- Words which make up sentences, sentences which would make up paragraphs, paragraphs which make up chapters are some examples of syntagms and syntagmatic relationships.
- To be more clear, words in a sentence can be considered as syntagms, and they form a syntagmatic relationship that gives rise to meaning.
- Changing the sequence of syntagms in the sentence can result in a change of meaning.

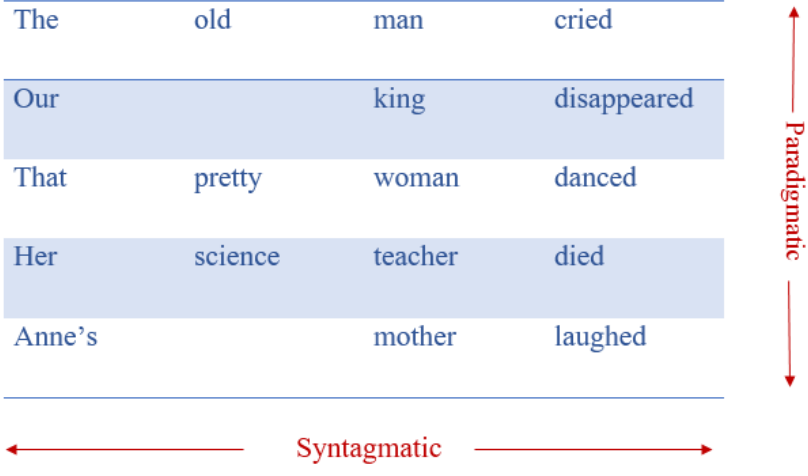
What is Syntagm?

- For example,
Anne killed a mosquito.
A mosquito killed Anne.
- The above sentences use the same words (syntagms), but the difference in the order (syntagmatic relationship) give two very different meanings.

Paradigm vs. Syntagm

- The following table might help you to understand the difference between paradigm and syntagm and their relationship more clearly.
- The syntagmatic relationship can be observed along the horizontal axis, and the paradigmatic relationship can be observed along the vertical axis.

The	old	man	cried
Our		king	disappeared
That	pretty	woman	danced
Her	science	teacher	died
Anne's		mother	laughed



Form and Function

- An understanding of form and function is a variation on the "knowing what vs. knowing how" discussion:
 - Form refers to the name of a thing (along with its definition)
 - Function refers to how a thing is acting or operating in a particular context.
- When analyzing grammar in this textbook, understanding the difference between FORM and FUNCTION is one of the keys to successful analysis of words, phrases, clauses, and sentence in their context.

Form and Function

- FORM names things at the WORD level (noun, verb, adjective, adverb, preposition, relative, etc.), at the PHRASE level (main verb phrase, prepositional phrase, present participle phrase, past participle phrase, gerund phrase, infinitive phrase, etc.), and at the CLAUSE level (independent clause, dependent clause, subordinate clause, relative clause, nominal clause, etc.).
- There are lots of FORMS, but a finite number of FUNCTIONS.

Functions

- In fact, there are only four (4) FUNCTIONS:
- nominal - a noun or any group of words that can substitute for a noun (word, phrase, or clause) and perform nominal functions. In other words, they can do anything that a noun can do, including filling any of the common nominal slots in a sentence (subject, direct object, subject complement, or object of the preposition most commonly).
- adjectival - an adjective or any group of words that can substitute for an adjective (word, phrase, or clause) and perform adjectival functions. In other words, they can do anything that an adjective can do (modify nouns or nominals most commonly).

Functions

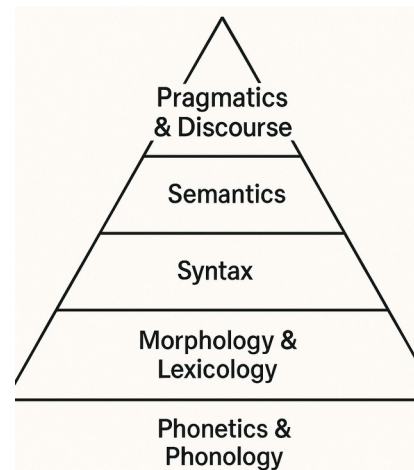
- adverbial - an adverb or any group of words that can substitute for an adverb (word, phrase, or clause) and perform adverbial functions. In other words, they can do anything that an adverb can do.
- main verb phrase (MVP) - the main verb of a clause plus its auxiliary or helping verbs in an unchanging order.

Analysis Example: "I saw her duck"

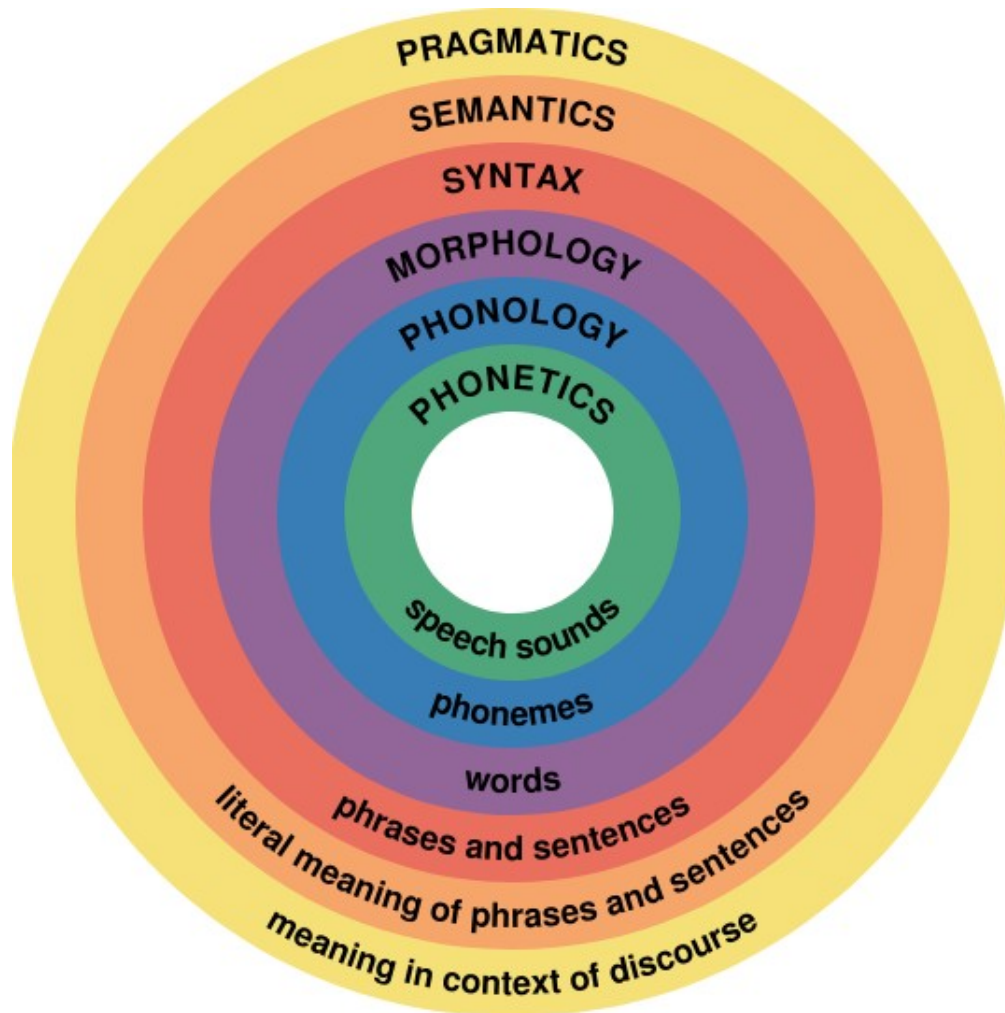
- This sentence has one Form (one string of words) but two completely different analyses.
-
-
-
-
- Takeaway: This is why PoS tagging (finding the function) is the first step to resolving semantic meaning.

The "Stack" of Language Analysis

- Language is not one single system; it's a "stack" of systems, from concrete to abstract.
- Analysis is done in layers, where each level builds on the one below it.
- This "stack" provides a clear roadmap for building an NLP pipeline.



Linguistic Analysis



Level 1: Phonetics (The Physics of Sound)

- Question: How are human speech sounds produced and perceived?
- It's the physics and biology of speech (e.g., "articulatory phonetics" - what your tongue and lips are doing).
- Example: The [p] in "pill" (aspirated, [p^h]) is physically different from the [p] in "spill" (unaspirated, [p]).
- NLP Application: The core of Speech-to-Text (ASR) and Text-to-Speech (TTS). Analyzing spectrograms to identify acoustic features.

Level 2: Phonology (The System of Sound)

- Question: How do sounds function as a system in a specific language?
- It's the abstract rules of sound.
- Example: In English, [p] and [p^h] are the same "phoneme" (/p/). But in Marathi, /p/ (प) and /p^h/ (फ) are different phonemes that change a word's meaning.
- Phonology also defines valid sound sequences (e.g., "brick" is valid, "bnick" is not).
- NLP Application: Language Models for ASR (knowing "bnick" is unlikely), morphological analysis.

Phonetics

VOWELS	monophthongs				diphthongs		PHONETIC CHART voiced unvoiced	
	i:	ɪ	ʊ	u:	ɪə	eɪ		
	Need	ship	good	Few	here	Pay		
	e	ə	ɜ:	ɔ:	ʊə	ɔɪ		
	Went	Alive	bird	door	tourist	noise	əʊ	show
	æ	ʌ	ɑ:	ɒ	eə	aɪ	aʊ	cow
	cat	Fun	far	Rob	care	my		
CONSONANTS	p	b	t	d	tʃ	dʒ	k	g
	pin	bag	tea	dog	chose	June	cash	go
	f	v	θ	ð	s	z	ʃ	ʒ
	fly	vest	think	Those	see	zoo	Shirt	treasure
	m	n	ŋ	h	l	r	w	j
	man	now	bang	hat	law	red	wall	yes

Level 3: Morphology (The Structure of Words)

- Question: How are words built from smaller, meaningful units?
- These units are called Morphemes.
- Example: un-believe-able
 - un-: (Bound Morpheme) "not"
 - believe: (Free Morpheme) the root concept
 - -able: (Bound Morpheme) "can be"
- NLP Application: Stemming (crude: running -> run) and Lemmatization (smart: was -> be). Crucial for search engines and reducing data sparsity.

Level 4: Lexicology (The Word Stock)

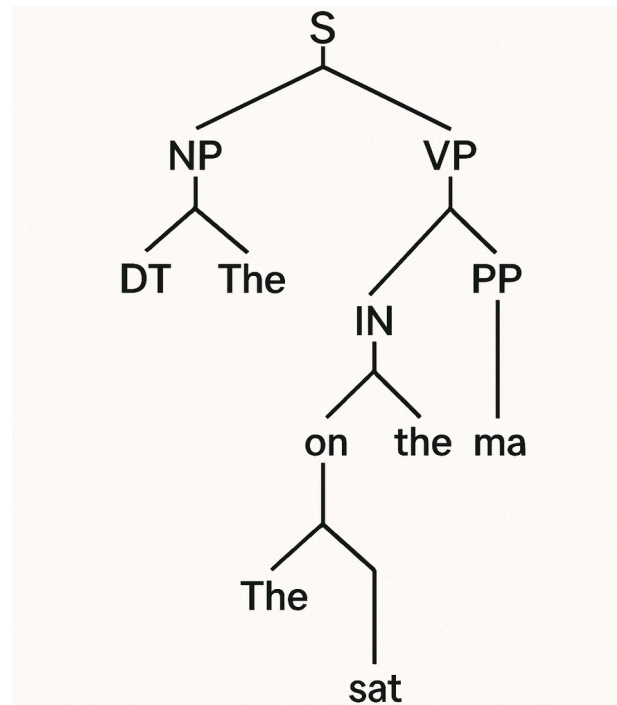
- Question: What is the relationship between words in a language's dictionary (the "lexicon")?
- It's the study of the word "inventory" of a language.
- It deals with:
 - Lexical Semantics: What does "bank" mean?
 - Relationships: (synonyms: big/large), (antonyms: hot/cold), (hyponyms: dog/animal).
- NLP Application: WordNet, Word Sense Disambiguation (WSD), creating knowledge graphs, and evaluating word embeddings.

Level 5: Syntax (The Structure of Sentences)

- Question: How are words combined to form valid, grammatical sentences?
- These are the rules of grammar (e.g., Subject-Verb-Object in English).
- The cat chased the dog. (Valid SVO)
- *Chased the cat the dog. (Invalid - * marks ungrammaticality)
- NLP Application: Part-of-Speech (PoS) Tagging, Chunking (finding phrases), and Dependency/Constituency Parsing (creating syntax trees).

Visualizing Syntax: The Parse Tree

- A parse tree is a data structure that explicitly represents the syntactic structure of a sentence, based on a formal grammar.



Level 6: Semantics (The Literal Meaning)

- Question: What does a sentence literally mean?
- It's the "truth-conditional" meaning, independent of context.
- Compositional Semantics: The meaning of a sentence is built from the meaning of its words (lexicology) and its structure (syntax).
- The cat chased the dog. vs. The dog chased the cat.
- These have the same words, but syntax changes the roles (Agent, Patient) and thus the literal meaning.
- NLP Application: Sentiment Analysis, Logical Inference, and all "understanding" tasks.

Level 7: Pragmatics (The Intended Meaning)

- Question: What did the speaker mean by saying that, in this specific context?
- It's the "meaning in context" or the "meaning between the lines."
- Example:
 - Speaker A: "Is it raining outside?"
 - Speaker B: "I just saw someone walk in with a wet umbrella."
 - Semantics (Literal): Someone has a wet umbrella.
 - Pragmatics (Intended): "Yes, it is raining."
- NLP Application: The hardest part. Chatbots, Dialogue Agents, and Summarization.

Level 8: Discourse (The Meaning Across Sentences)

- Question: How do multiple sentences combine to form a coherent, unified "text"?
- It's the analysis of "text-level" structure.
- Anaphora Resolution:
 - "I saw a cat on the mat. It was black."
 - Discourse analysis connects "It" back to "cat".
- Coherence: Why does "A then B" make sense, but "A then Z" doesn't?
- NLP Application: Text Summarization (finding the main topic), Question Answering (finding the answer in a paragraph), and Chatbot memory.

Linguistics -> NLP Pipeline

Linguistic Level	Core NLP Task
Phonetics/Phonology	Speech-to-Text (ASR)
Morphology	Lemmatization, Stemming
Lexicology	Word Sense Disambiguation (WSD)
Syntax	PoS Tagging, Parsing, NER
Semantics	Sentiment Analysis, NLI
Pragmatics	Dialogue Systems, Intent Recognition
Discourse	Summarization, Anaphora Resolution

Where Do We Fit?

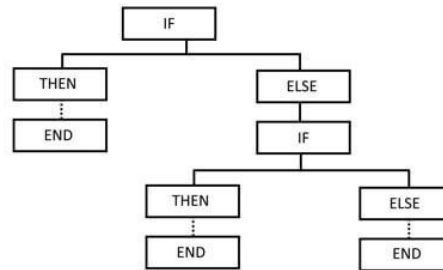
- Artificial Intelligence (AI): The broad goal of creating "intelligent" machines.
- Machine Learning (ML): A sub-discipline of AI where systems learn from data rather than being explicitly programmed.
- Deep Learning (DL): A technique within ML that uses neural networks with many layers.
- NLP (Our Field): A domain of AI that focuses on language. It uses ML and DL to solve its problems.

Types

- “Strong” Artificial Intelligence
 - Computers thinking at a level that meets or surpasses people
 - Computers engaging in abstract reasoning & thinking
 - This is not what we have today
 - There is no evidence that we are close to Strong AI
- “Weak” Pattern-Based Artificial Intelligence
 - Computers solve problems by detecting useful patterns
 - Pattern-based AI is an Extremely powerful tool
 - Has been used to automate many processes today
 - Driving, language translation
 - This is the dominant mode of AI today

Major AI Approaches

- Two Major AI Techniques
 - Logic and Rules-Based Approach



- Machine Learning (Pattern-Based Approach)

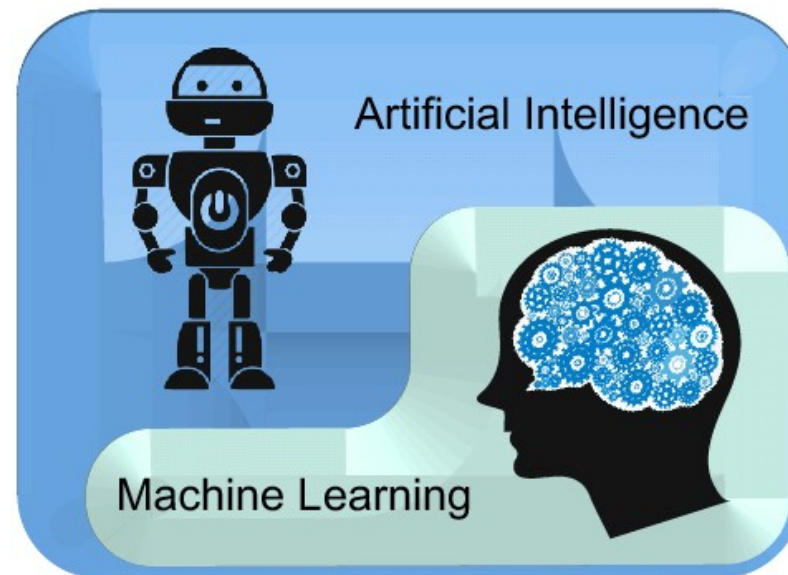


Rules based approach

- Logic and Rules-Based Approach
 - Representing processes or systems using logical rules
 - Top-down rules are created for computer
 - Computers reason about those rules
 - Can be used to automate processes

Learning vs. Designing

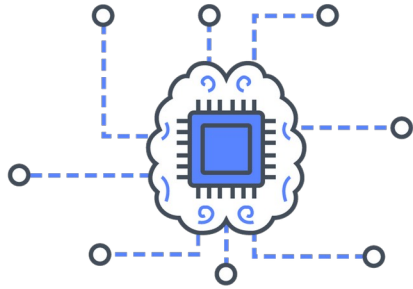
- AI is a bigger concept to **design** intelligent machines that can simulate human thinking capability and behavior, whereas, machine learning is an application or subset of AI that allows machines to **learn** from data without being programmed explicitly.



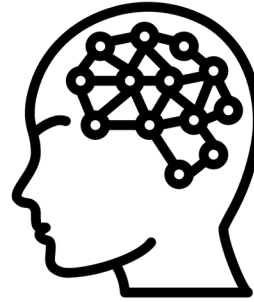
Branches Of Artificial Intelligence

- Artificial Intelligence can be used to solve real-world problems by implementing the following processes/ techniques:
 - Machine Learning
 - Deep Learning
 - Natural Language Processing
 - Robotics
 - Expert Systems
 - Fuzzy Logic

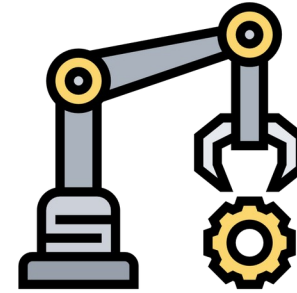
Branches Of Artificial Intelligence



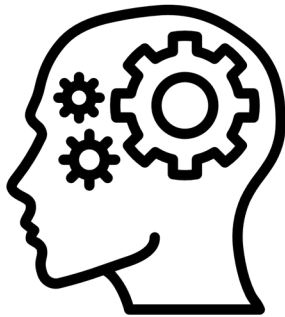
Machine Learning



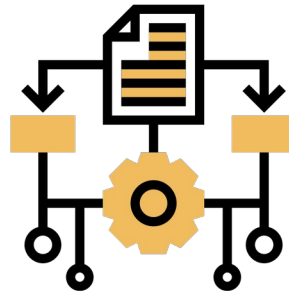
Neural Networks



Robotics



Expert Systems



Fuzzy Logic



Natural Language
Processing

Machine Learning

- Machine learning is an application of **artificial intelligence** (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed. Machine learning focuses on the development of computer programs that can access data and use it learn for themselves.
- The process of learning begins with **observations** or data, such as examples, **direct experience**, or **instruction**, in order to look for patterns in data and make better decisions in the future based on the examples that we provide.
- The primary aim is to allow the computers learn automatically **without** human intervention or assistance and adjust actions accordingly.

Machine Learning

Traditional Programming

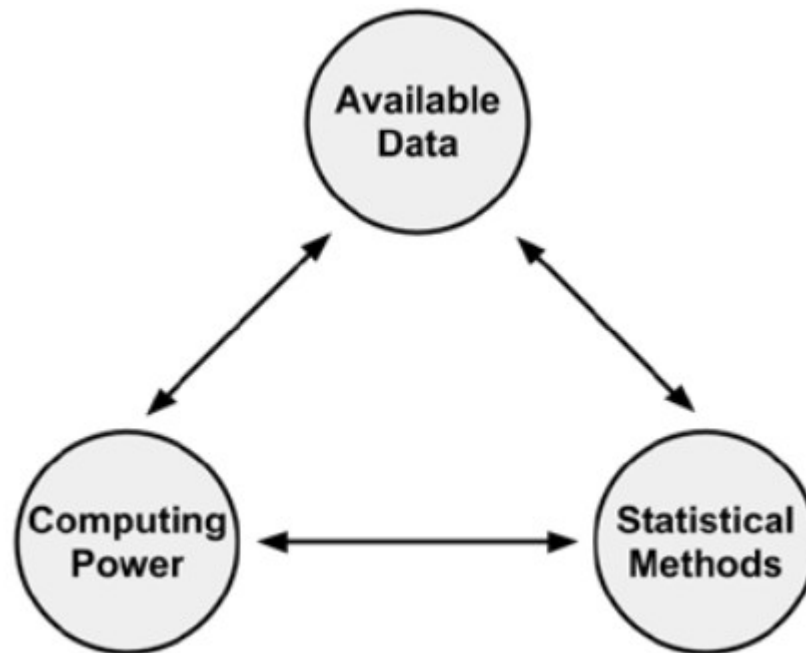


Machine Learning



Machine Learning

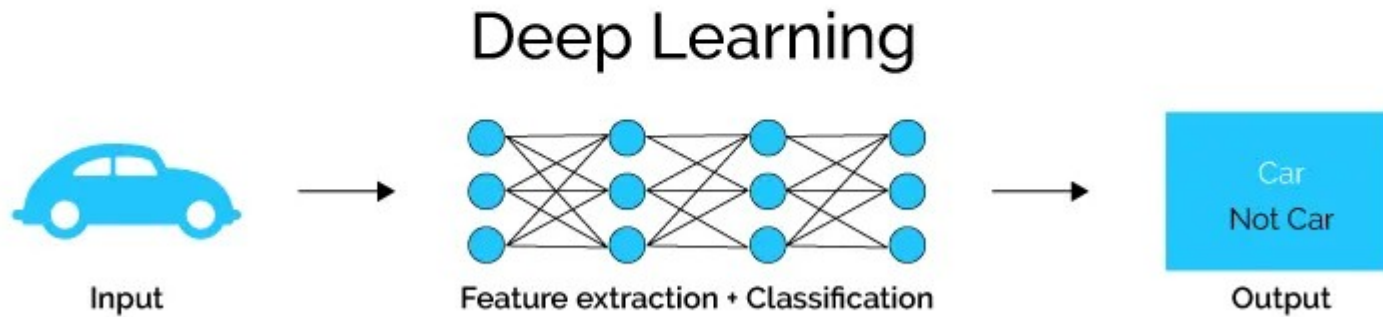
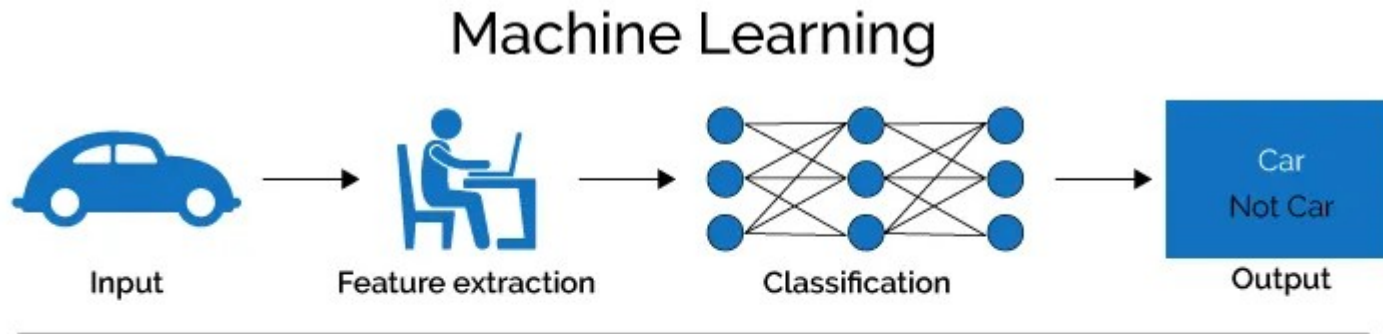
- The field of study interested in the development of computer algorithms for transforming data into intelligent action is known as machine learning.



Deep Learning

- Deep learning is a subset of machine learning, which is essentially a neural network with three or more layers.
- These neural networks attempt to simulate the behavior of the human brain—albeit far from matching its ability—allowing it to “learn” from large amounts of data.
- While a neural network with a single layer can still make approximate predictions, additional hidden layers can help to optimize and refine for accuracy.

Deep Learning



Robotics

- Robotics is a branch of engineering that involves the conception, design, manufacture and operation of robots.
- The objective of the robotics field is to create intelligent machines that can assist humans in a variety of ways.
- Robotics can take on a number of forms. A robot may resemble a human, or it may be in the form of a robotic application, such as robotic process automation (RPA), which simulates how humans engage with software to perform repetitive, rules-based tasks.

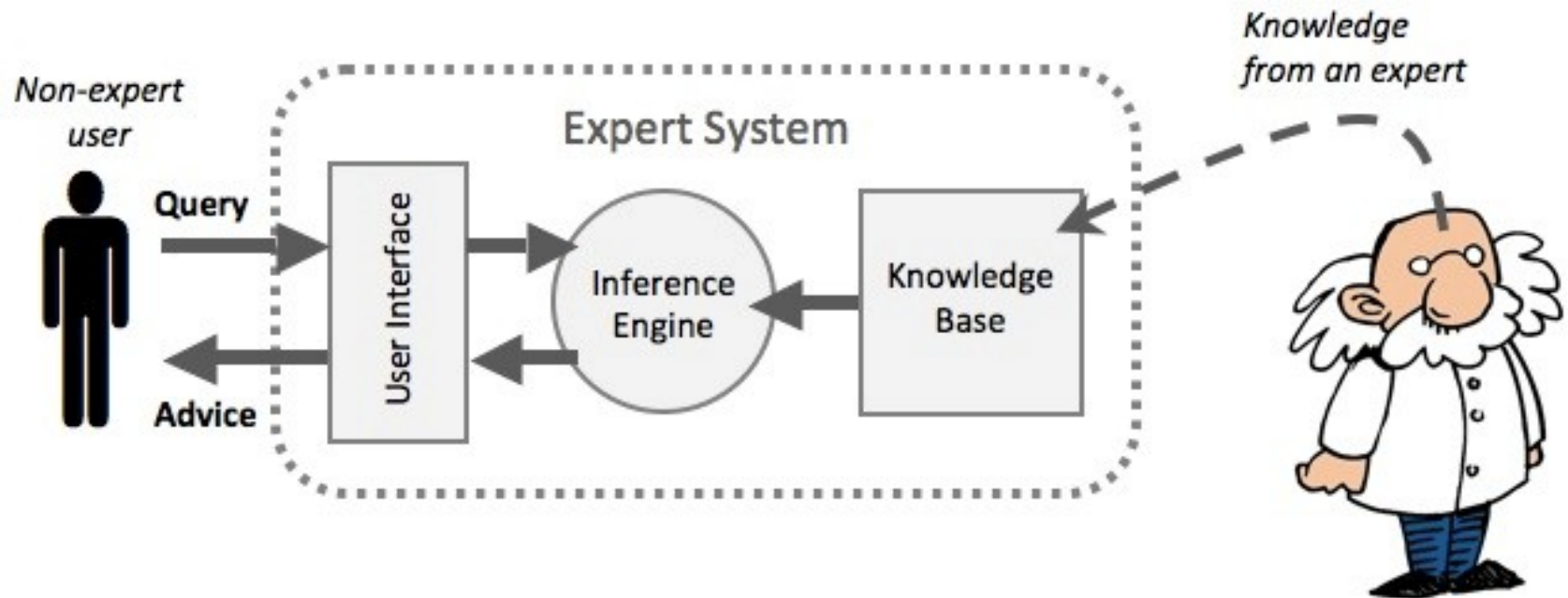
Robotics



Expert Systems

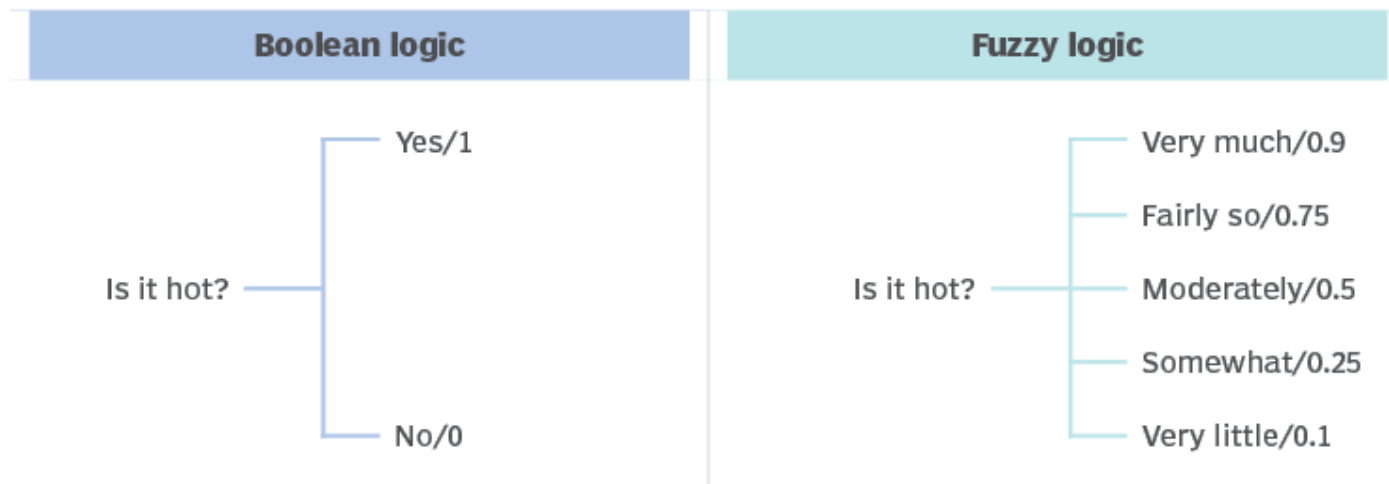
- An expert system is a computer program that uses artificial intelligence (AI) technologies to simulate the judgment and behavior of a human or an organization that has expertise and experience in a particular field.
- Expert systems are usually intended to complement, not replace, human experts.
- Expert systems accumulate experience and facts in a knowledge base and integrate them with an inference or rules engine -- a set of rules for applying the knowledge base to situations provided to the program.

Expert Systems



Fuzzy Logic

- Fuzzy logic is an approach to computing based on "degrees of truth" rather than the usual "true or false" (1 or 0) Boolean logic on which the modern computer is based.



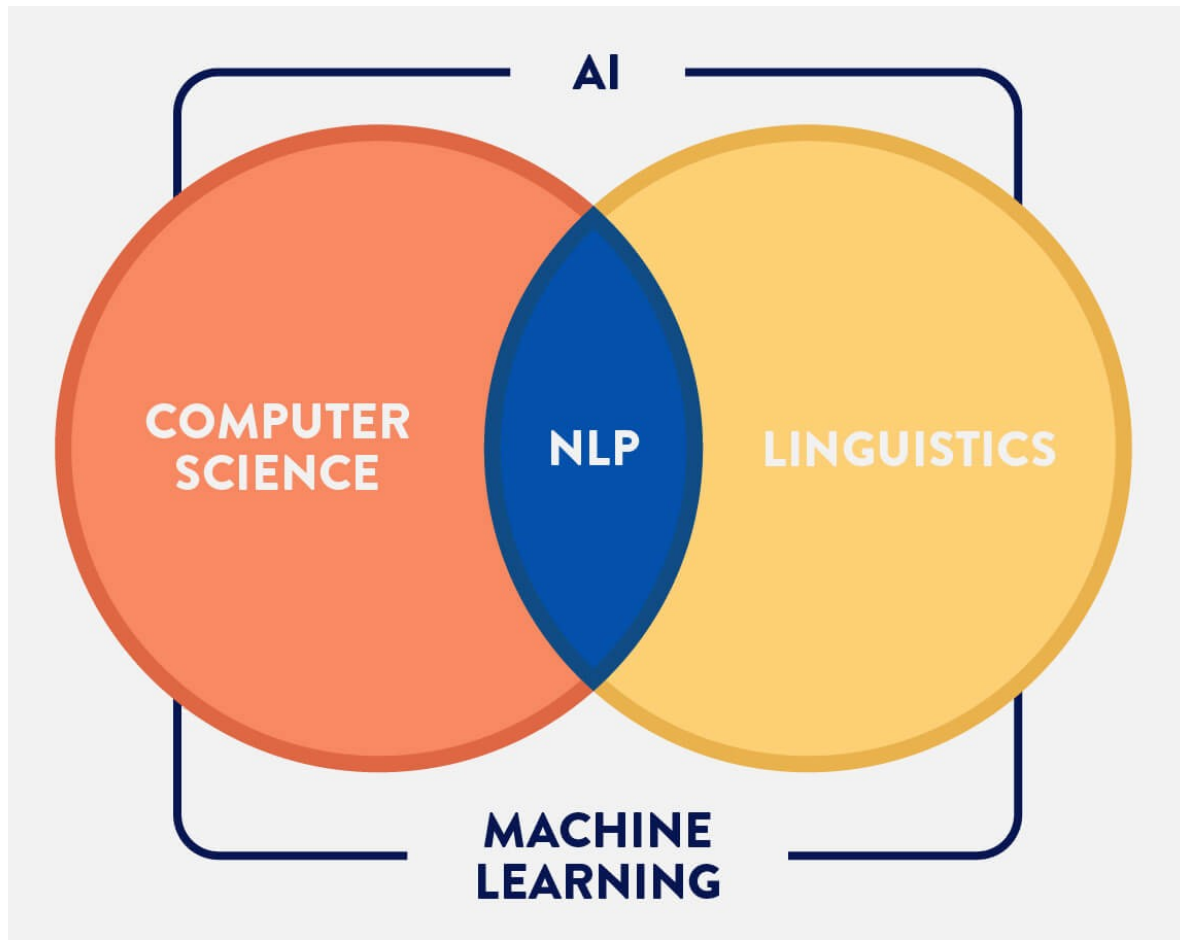
Natural Language Processing

- Natural language processing (NLP) is the ability of a computer program to understand human language as it is spoken and written -- referred to as natural language. It is a component of artificial intelligence (AI).
- NLP has existed for more than 50 years and has roots in the field of linguistics.
- It has a variety of real-world applications in a number of fields, including medical research, search engines and business intelligence.

NLP vs. Computational Linguistics

- These terms are often used interchangeably, but there's a subtle, important difference:
- Computational Linguistics (CL):
 - More scientific / academic.
 - Goal: Use computation to understand language (e.g., build a model to test a syntactic theory).
- Natural Language Processing (NLP):
 - More engineering / applied.
 - Goal: Use language to build useful applications (e.g., build a spam filter or a translator).

Natural Language Processing



The Pillars of NLP

- Almost every NLP task falls into one of three categories:
- 1. Natural Language Understanding (NLU): Reading / Listening
- 2. Natural Language Generation (NLG): Writing / Speaking
- 3. Natural Language Interaction (NLI): Conversing

NLU: Deconstructing to Meaning

- Goal: To convert ambiguous, unstructured Natural Language into an unambiguous, structured representation of meaning.
- It is a process of analysis and deconstruction.
- It's the "input" side of the problem.
- Question: What does the user mean?

NLU Task Examples

- Intent Recognition:
 - Text: "Book me a flight to Delhi" -> Intent: book_flight, Destination: Delhi
- Named Entity Recognition (NER):
 - Text: "Apple is launching a new iPhone in Cupertino." -> ORG: Apple, PRODUCT: iPhone, LOC: Cupertino
- Sentiment Analysis:
 - Text: "This movie was... not great." -> Sentiment: Negative
- Part-of-Speech (PoS) Tagging & Parsing: (As seen before)

NLU is the Foundation

- You cannot act on an instruction you don't understand.
- NLU is the critical first step for almost all intelligent systems:
 - Search Engines (Google)
 - Virtual Assistants (Siri, Alexa)
 - Data Analytics (sentiment tracking)
 - Machine Translation

NLG: Constructing from Meaning

- Goal: To convert structured data or a "meaning representation" into human-readable, fluent, and correct Natural Language.
- It is a process of synthesis and construction.
- It's the "output" side of the problem.
- Question: How do I express this concept?

NLG Task Examples

- Template-Based NLG (Simple):
 - Data: {Temp: 25C, Cond: Sunny} -> "Today's weather is 25C and Sunny."
- Abstractive Summarization:
 - Input: [10-page document] -> Output: [Two-paragraph summary]
- Machine Translation (MT):
 - Input: [French text] -> Output: [English text]
- Creative Generation:
 - Input: [Prompt] -> Output: [Poem, story, code] (e.g., GPT-3, BERT)

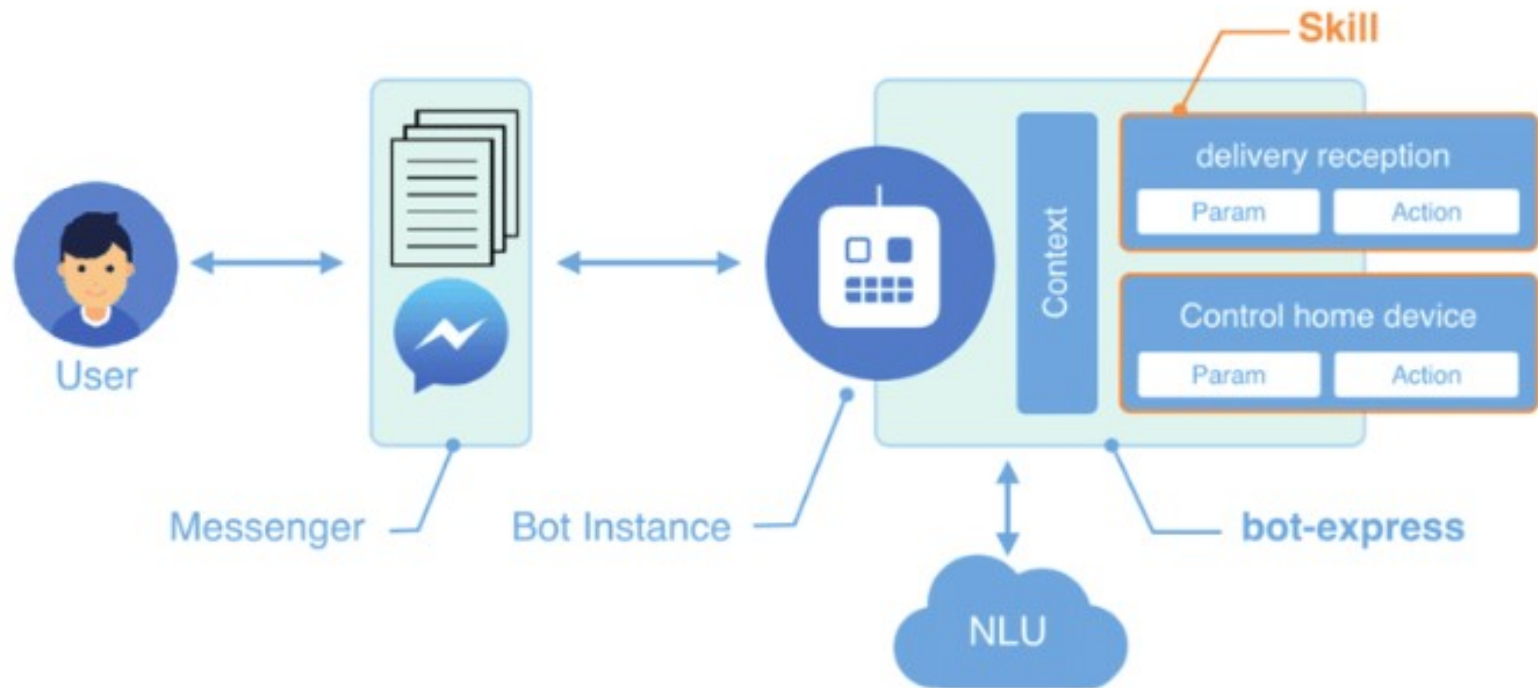
How NLG Works

- NLG is often a 3-stage pipeline (though Deep Learning blurs these lines):
 - 1. Content Determination: What to say? (e.g., select the most important data points).
 - 2. Sentence Planning: How to organize it? (e.g., order the facts, choose words like "but" or "and").
 - 3. Surface Realization: Write the final sentence. (e.g., apply grammar rules, is/are, a/an).
- Modern Transformer models (like GPT) try to do all three at once, "end-to-end".

NLI: The "Holy Grail"

- Goal: To build systems that can have a coherent, multi-turn, and useful conversation with a human.
- It is NOT just one task. It is the complete, end-to-end integration of:
 - NLU (to understand the user's turn)
 - Dialogue Management (to track the conversation, manage state, and decide what to do next)
 - NLG (to generate the system's response)

Natural Language Understanding



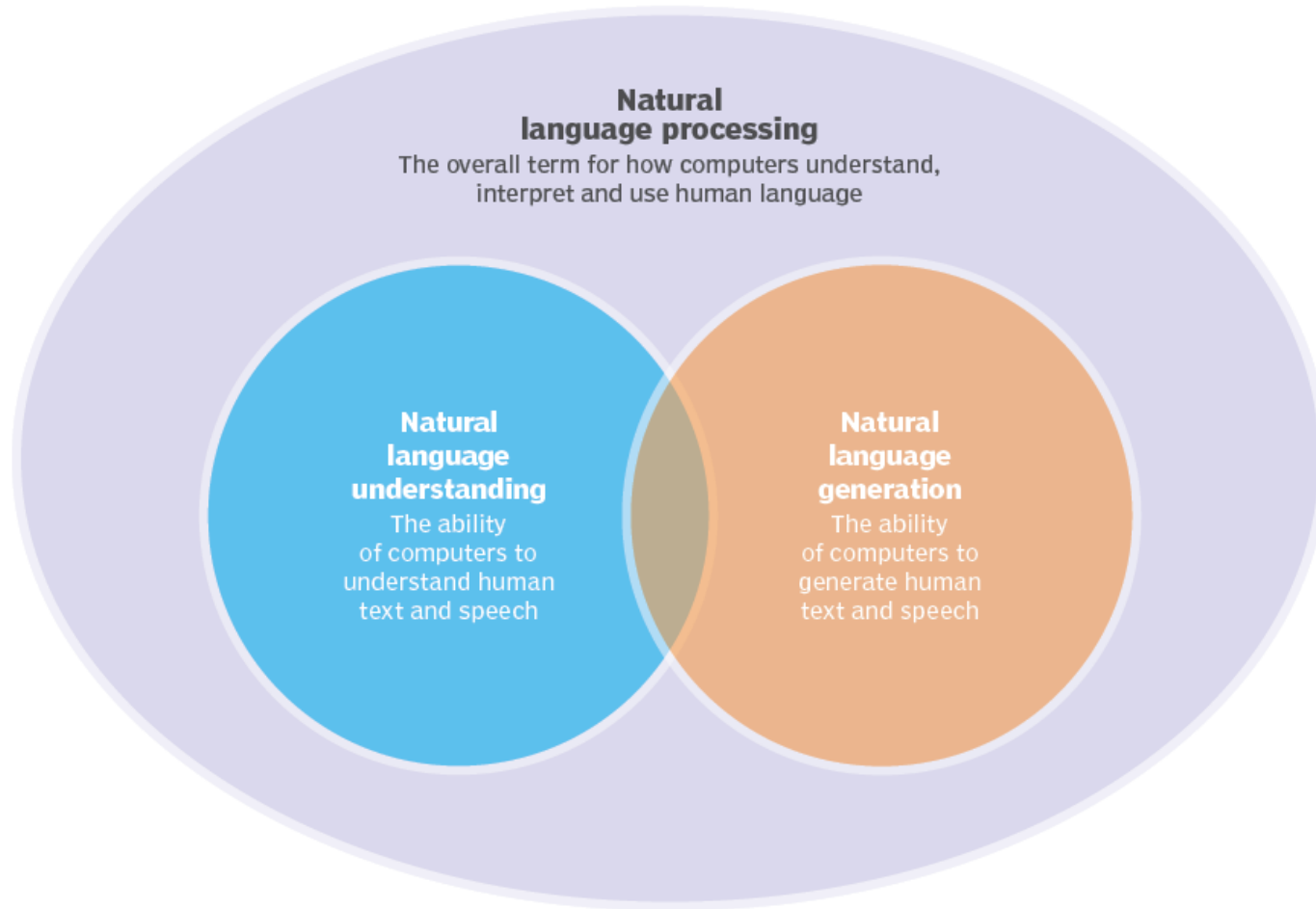
NLI in Action

- User: "I want to book a flight." (NLU: Intent=book_flight)
- Bot: "Sure, where are you flying to?" (DM: 'destination' is missing. NLG: Ask for 'destination')
- User: "Delhi." (NLU: Intent=inform_destination, Dest=Delhi)
- Bot: "And where are you flying from?" (DM: 'origin' is missing. NLG: Ask for 'origin')
- User: "From Pune, on Saturday." (NLU: Intent=inform_origin+date, Origin=Pune, Date=...)
- Bot: "OK. Looking for flights from Pune to Delhi for Saturday..." (DM: All slots filled. NLG: Confirm + execute)
- This state-tracking and interaction is the core of NLI.

The Frontier of AI

- NLI is the technology behind all major consumer-facing AI.
- Examples:
 - Virtual Assistants (Siri, Google Assistant, Alexa)
 - Customer Service Bots (on websites)
 - Generative AI (ChatGPT, Claude)
- The challenge is moving from simple "task-oriented" bots (like booking a flight) to complex, "knowledge-oriented" conversational partners.

Relations



NLG Models and Methodologies

- NLG relies on machine learning algorithms and other approaches to create machine-generated text in response to user inputs. Some of the methodologies used include the following:
- Markov chain.
 - The Markov model is a mathematical method used in statistics and machine learning to model and analyze systems that are able to make random choices, such as language generation.
 - Markov chains start with an initial state and then randomly generate subsequent states based on the prior one.

NLG Models and Methodologies

- Recurrent neural network (RNN).
 - These AI systems are used to process sequential data in different ways.
 - RNNs can be used to transfer information from one system to another, such as translating sentences written in one language to another.
 - RNNs are also used to identify patterns in data which can help in identifying images.
 - An RNN can be trained to recognize different objects in an image or to identify the various parts of speech in a sentence.

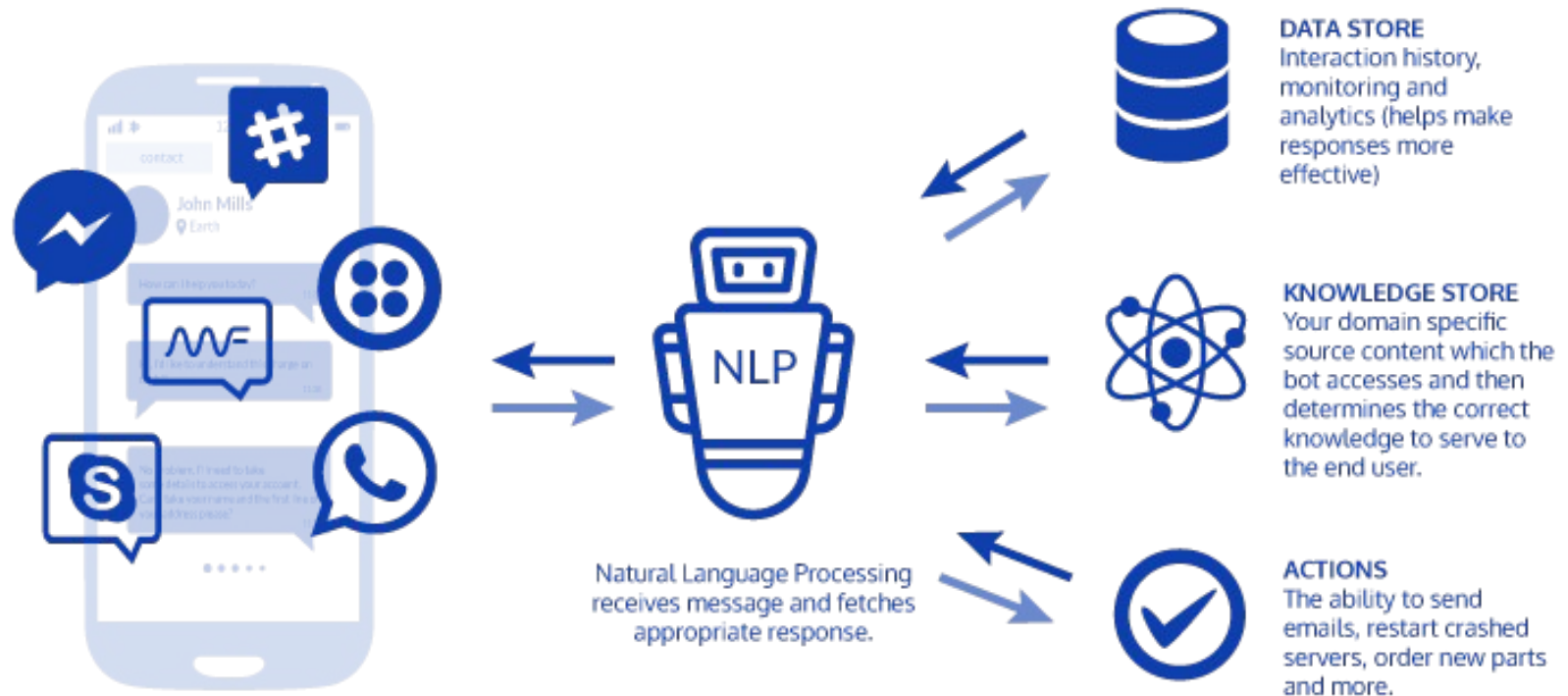
NLG Models and Methodologies

- Long short-term memory (LSTM).
 - This type of RNN is used in deep learning where a system needs to learn from experience.
 - LSTM networks are commonly used in NLP tasks because they can learn the context required for processing sequences of data.
 - To learn long-term dependencies, LSTM networks use a gating mechanism to limit the number of previous steps that can affect the current step.

NLG Models and Methodologies

- Transformer.
 - This neural network architecture is able to learn long-range dependencies in language and can create sentences from the meanings of words.
 - Transformer is related to AI. It was developed by OpenAI, a nonprofit AI research company in San Francisco.
 - Transformer includes two encoders: one for processing inputs of any length and another to output the generated sentences.

Summary



Computational Linguistics

- Computational linguistics (CL) is the application of computer science to the analysis and comprehension of written and spoken language.
- As an interdisciplinary field, CL combines linguistics with computer science and artificial intelligence (AI) and is concerned with understanding language from a computational perspective.
- Computers that are linguistically competent help facilitate human interaction with machines and software.

Computational Linguistics

- A computational linguist is required to have expertise in machine learning (ML), deep learning, AI, cognitive computing and neuroscience.
- Individuals pursuing a job as a linguist generally need a master's or doctoral degree in a computer science-related field or a bachelor's degree with work experience developing natural language software.
- The term computational linguistics is also very closely linked to NLP, and these two terms are often used interchangeably.

Computational Linguistics: Applns



Machine translation



Application clustering



Sentiment analysis



Chatbots



Knowledge extraction



Natural language interfaces



Content filtering

Computational Linguistics: Goals

- Create grammatical and semantic frameworks for characterizing languages.
- Translate text from one language to another.
- Retrieve text that relates to a specific topic.
- Analyze text or spoken language for context, sentiment or other affective qualities.
- Answer questions, including those that require inference and descriptive or discursive answers.
- Summarize text.
- Build dialogue agents capable of completing complex tasks such as making a purchase, planning a trip or scheduling maintenance.
- Create chatbots capable of passing the Turing Test.

Summary

- 1. Analysis is Deconstruction: We analyze language on two axes (Syntagmatic/Chain and Paradigmatic/Choice) and through three lenses (Form, Function, Meaning).
- 2. Analysis is a "Stack": We build up from sounds (Phonology) to words (Morphology) to sentences (Syntax) to literal meaning (Semantics) to intended meaning (Pragmatics & Discourse).
- 3. NLP is the "How": Each linguistic level maps directly to a core NLP task (e.g., Syntax -> Parsing).

Summary

- 4. NLP is a field of AI that uses ML and DL to solve language problems.
- 5. NLU (Understanding): Deconstructs language into data (e.g., Intent Recognition, NER).
- 6. NLG (Generation): Constructs language from data (e.g., Summarization, Translation).
- 7. NLI (Interaction): The complete, stateful loop that combines NLU and NLG to create a "conversation" (e.g., Chatbots).

Thank you

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